

LLD Case Study 1: Distributed Job Scheduler

Target Patterns: Strategy Pattern, Command Pattern, Observer Pattern

Key Engineering Focus: Thread-safe task execution, pluggable scheduling algorithms, worker coordination, and lifecycle event streaming.

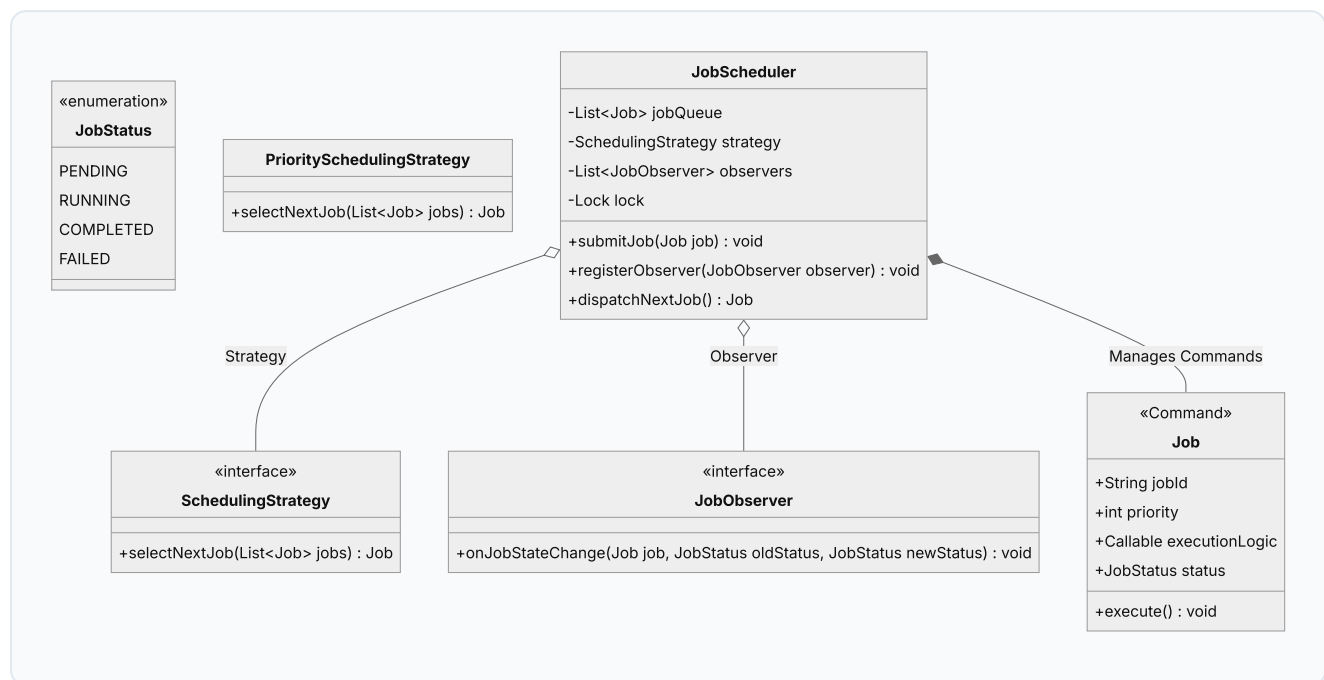
1. Problem Statement & Functional Requirements

Design a modular in-process Job Scheduler capable of scheduling, queuing, executing, and monitoring tasks across worker pools.

Requirements:

- Job Definition (Command Pattern):** Encapsulate jobs as executable units of work with retry counts and parameters.
 - Scheduling Policy (Strategy Pattern):** Support pluggable scheduling algorithms (e.g. First-Come-First-Served / Priority Scheduling vs Shortest Job First).
 - Lifecycle Notifications (Observer Pattern):** Allow external monitoring systems (logging, metrics, alerting) to subscribe to job state changes (`PENDING` , `RUNNING` , `COMPLETED` , `FAILED`).
 - Thread-Safety:** Multiple threads must be able to submit and execute jobs concurrently without state corruption.
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2. Architecture & UML Class Diagram



3. Production-Grade Python Implementation